

Fiber Cancellation Creates a QFT Certification Gap in Finite Regev-Style Factoring

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Abstract

Distance-truncated quantum Fourier transforms (QFTs) remove small controlled rotations, but a safe transform-level error bound can be much more demanding than a factoring algorithm’s actual needs. We audit one such bound in a finite Regev-style pipeline and prove that it cannot certify any non-exact cutoff when $M < 4\pi dm/\Delta$. This is a limitation of the certificate, not a lower bound on useful Fourier precision. Modular-exponentiation fibers can cancel coherent phase error before measurement, and an explicit finite case has the same exact and truncated measurement law although the certificate rejects. We then freeze eight small semiprimes, two exact finite input-state models, three Fourier moduli, all cutoffs, and a complete sample-to-augmented-lattice-to-LLL-to-factor endpoint. The certificate allows zero omitted layers, whereas one omitted layer meets a preregistered 10-percentage-point non-inferiority rule in all six model-modulus cells; two layers pass for the finite-Gaussian model at $M = 16, 32$. The largest passing cutoff removes six logical controlled phases and twelve QFT-only transpiled CX gates. These results establish a finite, model- and decoder-specific certification gap, not asymptotic approximate-QFT sufficiency or end-to-end hardware speedup.

1 Introduction

Shor’s algorithm made integer factoring a central application of quantum period finding [1, 2]. Regev later proposed a factoring algorithm that reduces the quantum gate count per circuit, under a number-theoretic conjecture, by using multidimensional Fourier samples and more substantial classical lattice reduction [3]. This trade raises a practical precision question: must every controlled phase in each Fourier register be retained for the downstream lattice decoder to work?

Approximate QFTs are not new. Prior work analyzes truncated or banded QFTs inside factoring and period-finding algorithms, including their task-level success rather than only matrix error [4, 5, 6, 7, 8]. We therefore ask a narrow question specific to the audited

repository:

Does rejection by a conservative, all-input QFT certificate imply loss at a finite multidimensional Regev-style lattice-and-factor endpoint?

Our falsifiable hypothesis was that the certificate is conservative because it discards the modular-fiber and measurement structure of the prepared state. The study makes four bounded contributions.

1. It gives a line-by-line audit and exact scaling limitation for the implemented worst-case certificate.
2. It identifies arithmetic-fiber cancellation as a mechanism and provides an equal-measurement-law counterexample to the implication “certificate rejection means information loss.”
3. It evaluates every cutoff in a frozen eight-modulus holdout using a complete exact-lattice/LLL/stored-root factor endpoint, with the modulus—not repeated shot batches—as the unit of generalization.
4. It records a pre-holdout QFT-ordering correction, sharper finite factor-blind diagnostics, raw data, seeds, and resource counts.

The result is a *certification gap*: some rejected cutoffs tolerate a bounded empirical recovery loss in the tested regime. The truncated pipeline does not outperform exact-QFT recovery, and we do not claim a new QFT synthesis, a proof for full Regev sampling, or quantum advantage.

Implementation lineage. A companion artifact by Amorim, Barnes, and Malik developed an exact classical simulator for the uniform exponent-box Fourier law, retained the prime roots associated with squared circuit bases, implemented exact-rational LLL post-processing, and introduced a uniform-sample null control [9]. It has no finite-Gaussian input or approximate QFT circuit, so it is an engineering and methodological predecessor rather than evidence for the truncation claim here. The present repository adds the circuit audit, immutable

root provenance, explicit L versus L_0 classification, finite-Gaussian model, QFT interventions and certificates, and frozen held-out evaluation [10].

2 Finite Regev-style endpoint

2.1 Roots, squared bases, and useful relations

Let N be an odd composite, and choose units $b_1, \dots, b_d \pmod{N}$ without using the factors of N . The circuit bases are the paired residues

$$a_i = b_i^2 \pmod{N}. \quad (1)$$

The implementation stores every pair (b_i, a_i) immutably. This is necessary because a residue can have several modular square roots; factor extraction must use the root that generated the circuit base, not an arbitrary square root reconstructed later.

Define the arithmetic map and two relation lattices

$$h_A(x) = \prod_{i=1}^d a_i^{x_i} \pmod{N}, \quad (2)$$

$$L = \left\{ z \in \mathbb{Z}^d : \prod_i a_i^{z_i} = 1 \pmod{N} \right\}, \quad (3)$$

$$L_0 = \left\{ z \in L : \prod_i b_i^{z_i} \in \{1, -1\} \pmod{N} \right\}. \quad (4)$$

For $z \in L$, let $\beta(z) = \prod_i b_i^{z_i} \pmod{N}$. Then $\beta(z)^2 = 1 \pmod{N}$. A vector in L_0 is a valid but non-factor-yielding relation. A vector in $L \setminus L_0$ gives a nontrivial square root of one; the endpoint tests $\gcd(\beta - 1, N)$ and $\gcd(\beta + 1, N)$. It first verifies membership in L and performs this classification without known factors or group orders.

2.2 Exact finite sampling law

Let $M = 2^q$, $[M] = \{0, \dots, M-1\}$, and let $g(x) = \prod_i w(x_i)$ be a separable real amplitude. Before the product inverse QFT, the normalized joint state is

$$|\psi_w\rangle = \frac{1}{\sqrt{Z}} \sum_{x \in [M]^d} g(x) |x\rangle |h_A(x)\rangle, \quad Z = \sum_x |g(x)|^2. \quad (5)$$

After tracing out the arithmetic register, the exact finite measurement law for a cutoff- t inverse QFT $F_{M,t}^{-1}$ is

$$P_{A,w}^{(t)}(k) = \frac{1}{Z} \sum_y \left| \sum_{x: h_A(x)=y} g(x) \langle k | (F_{M,t}^{-1})^{\otimes d} | x \rangle \right|^2. \quad (6)$$

For the exact transform, $\langle k | F_M^{-1} | x \rangle = M^{-1/2} e^{-2\pi i k x / M}$. Equation (6) exposes the roots-of-unity sums inside each arithmetic fiber $\{x : h_A(x) = y\}$.

We evaluate two finite laws exactly up to floating-point roundoff. Model A is the notebook's uniform hard box, $w(j) = 1$. Model B uses centered truncated discrete-Gaussian amplitudes $w(j) = \exp(-\pi z_j^2 / R^2)$ with $z_j \in [-M/2, M/2)$ and $R = 4$. Model B is closer to Regev's state, but remains a small finite simulation rather than his asymptotic parameter regime.

2.3 Samples to an exact integer lattice

Given m measured rows $k_j \in [M]^d$, place them in $K \in \mathbb{Z}^{m \times d}$ and set $W = K/M$. With scale S , the augmented lattice is generated by the columns of

$$B = \begin{bmatrix} I_d & 0 \\ SW & SI_m \end{bmatrix}. \quad (7)$$

If $S = p_s / r_s$ in lowest terms, the code reduces the exactly cleared integer row basis

$$\mathcal{A} = (r_s M B)^\top = \begin{bmatrix} r_s M I_d & p_s K^\top \\ 0 & p_s M I_m \end{bmatrix}. \quad (8)$$

SymPy's exact integer LLL transform [11] exposes the candidate z in its first d transformation coefficients. The primary endpoint scans the Gram-Schmidt prefix specified by Regev's recovery argument, verifies $z \in L$, tests $z \in L_0$ using stored roots, and finally attempts the two gcds. No bounded enumeration, BKZ, quotient deflation, or known-factor oracle is used in the frozen endpoint.

3 Worst-case certificate and its gap

Cutoff $t \in \{0, \dots, q-1\}$ retains controlled phases whose qubit separation is at most t ; $t = q-1$ is exact. At separation r , a q -qubit QFT has $q-r$ phases of angle $\pi/2^r$. Thus the omitted-angle tail is

$$\eta(q, t) = \pi \sum_{r=t+1}^{q-1} \frac{q-r}{2^r}. \quad (9)$$

Proposition 1 (Audited certificate limitation). *For d Fourier registers, m independent executions, and allowed absolute event-probability change $0 < \Delta < 1$, the implemented certificate approves a cutoff only if*

$$\min(1, m \min(1, 2d\eta(q, t))) \leq \Delta. \quad (10)$$

Consequently,

$$M < \frac{4\pi dm}{\Delta} \implies \text{no non-exact cutoff is approved.} \quad (11)$$

Proof. A controlled phase obeys $\|\text{CP}(\theta) - I\|_2 = 2|\sin(\theta/2)| \leq |\theta|$. Telescoping the omitted gates gives $\|F_M - F_{M,t}\|_2 \leq \eta(q, t)$; telescoping the d tensor factors gives $\|F_M^{\otimes d} - F_{M,t}^{\otimes d}\|_2 \leq d\eta(q, t)$. The conservative

pure-state-to-measurement step yields one-shot TV $\leq \min(1, 2d\eta)$. Replacing one coordinate at a time in the m -sample product law gives Eq. (10), and every downstream event differs by at most this TV distance. The least aggressive non-exact cutoff is $t = q - 2$, for which $\eta(q, q - 2) = 2\pi/M$. Substitution, clipping, and $\Delta < 1$ give the strict implication in Eq. (11). $\square$

These inequalities are standard operator, data-processing, and hybrid bounds; we do not claim them as new. Their audited composition explains why the finite selector refuses all savings. For the frozen $d = 2, m = 7, \Delta = 0.05$ experiment, the right side of the threshold is about 3518.6, far above every tested $M \leq 32$.

3.1 Why rejection does not imply information loss

The operator norm maximizes over all input states, while Eq. (6) restricts amplitudes to modular fibers and then measures. Omitted phases can cancel within a fiber, alter only low-mass outcomes, or rotate amplitudes without changing their squared sums. For $N = 15$, squared bases $(4, 1)$, $d = 2$, $M = 4$, and $t = 0$, direct finite fiber summation gives identical exact and truncated measured laws (numerical TV $= 5.9 \times 10^{-35}$), although Eq. (10) rejects. This deliberately degenerate counterexample refutes a logical implication; it is not a broad factoring benchmark.

Three factor-blind finite refinements diagnose the gap: exact measured-law TV with an m -sample hybrid, exact product-Hellinger composition, and trace distance on the prepared joint fiber state. They are exponentially expensive in M^d and are validation tools, not scalable replacements for Regev’s analysis. In the proof-slack audit, the median omitted-gate triangle factor is 1.18, while the median state-to-measurement factor is 6.80. A separate $M = 128$ example nearly saturates the first gate step (ratio 0.9998996), locating the important conservatism after the gate-level bound.

Implementation correction. Before the holdout, matrix tests found that the custom approximate inverse QFT iterated qubits in the reverse gate order. The previous full-cutoff test had silently substituted a library QFT and therefore missed the error. The implementation was corrected so that retaining every phase reproduces the direct roots-of-unity matrix; sign, swaps, endianness, layer order, and each cutoff now have regression tests. Every pre-correction approximate-QFT row was invalidated and regenerated, and none enters this study.

4 Frozen experiment

The protocol was committed before the result manifest. It fixed roots $(2, 3)$ and their squared bases, $d = 2$, $M \in \{8, 16, 32\}$, $m = 7$, both models, every valid cutoff, $S = 4$,

relation bound $T = 4$, LLL parameter $3/4$, 64 coupled replicates per cell, and master seed 2026071301. Exact and approximate trials share seeds. The eight held-out semiprimes were

55, 65, 85, 95, 115, 119, 133, 161.

No modulus was excluded. Their known factors were used only after recovery to validate a returned pair; state construction, certificates, LLL, relation classification, and cutoff selection never receive them.

For each N , success probability is estimated over the 64 paired replicates. Inference resamples the eight whole- N clusters 5,000 times. A cutoff is declared empirically non-inferior only when the 95% lower confidence bounds for both factor recovery and verified $L \setminus L_0$ recovery are at least -0.10 relative to exact QFT. This ten-percentage-point engineering margin was frozen independently of the theorem’s stricter $\Delta = 0.05$ certificate budget. It is not an information-theoretic guarantee.

The experiment contains 192 per- N configurations and 12,288 raw trials. Its two primary outcomes are factor recovery and recovery of at least one verified $L \setminus L_0$ relation. These outcomes coincide in the selected cells because every recovered nontrivial stored-root class yielded the verified factor pair; this is observed, not assumed for arbitrary composites.

5 Results

The original certificate approved zero omitted layers in all six model-modulus cells. One omitted layer met the frozen non-inferiority rule in all six cells. Two layers passed only for finite-Gaussian Model B at $M = 16, 32$. Table 1 reports the largest passing omission in each cell. All recovery differences are approximate minus exact, and the means average the eight per- N probabilities.

The approximate pipeline does not improve recovery; it accepts a measured loss in exchange for fewer QFT gates. More aggressive cutoffs fail in most cells, so the result is not “truncation never matters.” The largest passing cutoff removes six logical controlled phases and twelve QFT-only transpiled CX gates. Depending on the cell, QFT-only depth savings range from zero to four. Modular arithmetic is not included and may dominate full-circuit depth, so these are not hardware-runtime or end-to-end savings.

Post-hoc margin sensitivity preserves at least one layer in all six cells at a 0.05 margin, but two layers remain only for Gaussian $M = 32$. At 0.02, hard-box $M = 8$ permits no omission. Thus the strongest “one layer in every cell” statement depends on tolerating more than two percentage points of absolute loss. Leave-one- N -out checks retain the primary decisions but do not create new held-out data.

A representative sharper-certificate row illustrates the mechanism. For $N = 55$, Model B, $M = 32$, and one

Table 1: Largest omission meeting the frozen rule. CI denotes the 95% whole- N cluster-bootstrap interval. Gate counts cover only the two product QFT blocks.

M	Model	Layers omitted	Exact mean	Approx. mean	Difference	95% CI	CP saved	CX saved
8	A	1	0.1191	0.1074	-0.0117	$[-0.0273, 0]$	2	4
8	B	1	0.0898	0.0879	-0.0020	$[-0.0117, 0.0059]$	2	4
16	A	1	0.1699	0.1641	-0.0059	$[-0.0137, 0]$	2	4
16	B	2	0.0840	0.0469	-0.0371	$[-0.0742, 0]$	6	12
32	A	1	0.1992	0.1934	-0.0059	$[-0.0176, 0]$	2	4
32	B	2	0.0703	0.0527	-0.0176	$[-0.0352, -0.0039]$	6	12

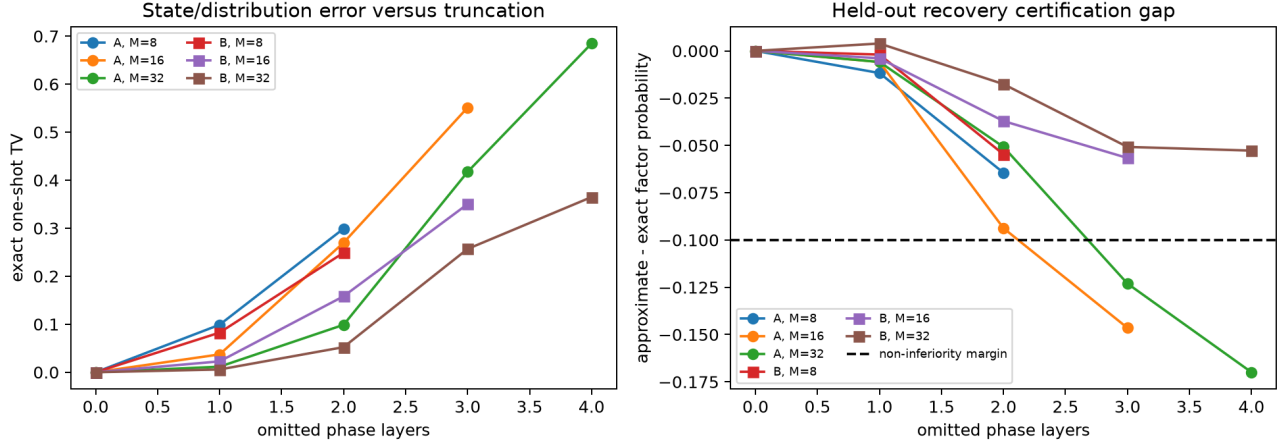


Figure 1: Left: exact one-shot distribution TV versus omitted layers for the representative modulus $N = 55$ only. Right: paired factor-probability differences aggregated across all eight held-out N values; the dashed line is the frozen -0.10 margin. Passing cutoffs lie above that line after the confidence-bound test, not merely by their plotted mean.

omitted layer, one-shot TV is 0.005748; the seven-shot TV and Hellinger bounds are 0.04024 and 0.03955, respectively. Both are below $\Delta = 0.05$ although the all-input certificate rejects. These finite refinements close only part of the empirical gap: the two-layer Gaussian cells are not uniformly certified across held-out moduli.

6 Prior-art boundary and limitations

Task-aware approximate-QFT evaluation is established in Shor-style period finding [4, 5, 6, 7, 8]; our claim is limited to the specified multidimensional sampler and exact Regev-style lattice endpoint. Recent work also studies implementations of Regev factoring [12] and noise-robust post-processing under a structured fraction of corrupted circuit executions [13]. Coherent QFT truncation changes every run’s distribution and is not automatically covered by the latter sparse-corruption model.

The holdout is small and structurally narrow: every N is a small semiprime containing factor 5 or 7. We test only $d = 2$, roots $(2, 3)$, $M \leq 32$, $m = 7$, $R = 4$, ideal finite A/B laws, one distance-based truncation, and one LLL decoder. Model A is a notebook surrogate. Model B is a truncated finite Gaussian, not a proof-level realization of Regev’s asymptotic state. There is no gate-level hard-

ware noise, calibrated device run, cryptographic-scale instance, full modular-arithmetic resource comparison, or proof that every postprocessor preserves recovery. With only eight N clusters, confidence intervals are necessarily coarse, and the frozen 0.10 margin is a design choice rather than a universal scientific threshold.

The exact state and finite certificate calculations scale exponentially in M^d . They certify and explain toy instances; they do not dequantize Regev’s algorithm in the complexity-theoretic sense. A scalable theorem exploiting fiber structure while bounding the complete lattice event remains open.

7 Conclusion

The experiments and audit support three separately scoped claims.

Verified implementation correction. The custom approximate-QFT order was wrong, the masking test was replaced, and all affected results were regenerated before evaluation.

Verified finite empirical result. For the frozen eight-semiprime A/B study and current LLL endpoint, the worst-case certificate allows zero truncation while the cut-

offs in Table 1 meet the declared recovery-loss criterion and save QFT-local gates.

Unverified general hypothesis. Arithmetic-fiber-aware reasoning may enable scalable precision selection for larger Regev instances. The present data do not establish that claim.

The practical lesson is modest but important: failure of a sufficient all-state certificate is not evidence of algorithmic or information-theoretic failure. Precision decisions should retain the structure of the prepared state, its measurement, and the actual classical endpoint whenever that structure can be analyzed.

Artifact and reproducibility

The accompanying archive contains this source, bibliography, frozen configuration, selected raw summary tables, the plotted figure, checksums, and a verification script. The complete code and all 12,288 trial rows are in the research repository at commit `a4a47c77aa9e5935597d9a3c37fc6efe9f8b1d04` [10]. The primary reproduction entry point is `scripts/run_qft_certificate_gap.py`.

Authorship note. The author block is provisional. Before external circulation or submission, the collaborators should confirm authorship, author order, affiliations, and which results may be publicly attributed.

References

- [1] Peter W. Shor. Algorithms for quantum computation: Discrete logarithms and factoring. In *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*, pages 124–134, 1994.
- [2] Peter W. Shor. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing*, 26(5):1484–1509, 1997.
- [3] Oded Regev. An efficient quantum factoring algorithm. *Journal of the ACM*, 72(1), 2025.
- [4] Don Coppersmith. An approximate fourier transform useful in quantum factoring. Technical Report RC 19642, IBM Research, 1994.
- [5] Adriano Barenco, Artur Ekert, Kalle-Antti Suominen, and Päivi Törmä. Approximate quantum fourier transform and decoherence. *Physical Review A*, 54(1):139–146, 1996.
- [6] Austin G. Fowler and Lloyd C. L. Hollenberg. Scalability of shor’s algorithm with a limited set of rotation gates. *Physical Review A*, 70(3):032329, 2004.
- [7] Y. S. Nam and R. Blümel. Performance scaling of shor’s algorithm with a banded quantum fourier transform. *Physical Review A*, 86(4):044303, 2012.
- [8] Y. S. Nam and R. Blümel. Scaling laws for shor’s algorithm with a banded quantum fourier transform. *Physical Review A*, 87(3):032333, 2013.
- [9] Carolina Amorim, Owen Barnes, and Summer Malik. Improvements to the regev factoring simulation. Software repository, 2026. Commit 595e755a93976ab865c795b61195b7f8c4fa7634.
- [10] Summer Malik and contributors. Regev-style circuit and lattice post-processing research artifact. Software repository, 2026. Research snapshot used for the paper: commit a4a47c77aa9e5935597d9a3c37fc6efe9f8b1d04.
- [11] Arjen K. Lenstra, Jr. Hendrik W. Lenstra, and László Lovász. Factoring polynomials with rational coefficients. *Mathematische Annalen*, 261(4):515–534, 1982.
- [12] Przemysław Pawlitko, Natalia Moćko, Marcin Niemiec, and Piotr Cholda. Implementation and analysis of regev’s quantum factorization algorithm. *Quantum Information & Computation*, 26(1):230–252, 2026.
- [13] Seyoon Ragavan and Vinod Vaikuntanathan. Space-efficient and noise-robust quantum factoring. *Journal of Cryptology*, 39(2), 2026.